

InsightAI: Human, AI, and Human–AI Co-Design of a Multimodal Big Data Decision Support System

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1. Problem Definition and Challenge 2 Goal

Overall Project Context: Modern enterprise e-commerce platforms ingest massive volumes of relational transactional data (orders, items, delivery tracking, payments) paired with unstructured customer feedback (free-text reviews and ratings). However, business executives, logistics managers, and end consumers struggle to convert this siloed data into actionable intelligence. Traditional Business Intelligence (BI) dashboards present numerical trends (*WHAT* happened) but cannot reveal root causes (*WHY* it happened). Conversely, unstructured text review analysis reveals complaints but lacks quantitative validation against actual delivery timestamps and carrier logs.

Target Users: The system serves two primary personas: (1) *E-Commerce Category Managers and Operations Analysts* who need to diagnose underperforming product segments, audit supplier bottlenecks, and trigger operational remediations; and (2) *Prospective Customers* seeking pre-purchase risk transparency regarding delivery delays and physical product defects.

Focused Challenge 2 Scope vs Overall Project: While the long-term semester vision encompasses multi-agent automated supplier negotiation and dynamic pricing optimization, Challenge 2 focuses on delivering one foundational, evidence-grounded vertical slice: **Natural Language Multimodal Business Data Analysis and Satisfaction Risk Prediction**. The system takes natural language questions regarding category performance or simulated order parameters, retrieves quantitative tabular metrics via SQL tools, retrieves qualitative review excerpts via semantic RAG, predicts satisfaction risk via a trained machine learning model, and structures the output under strict provenance with interactive human audit controls.

End-to-End Vertical Slice Workflow:

CHALLENGE 2 VERTICAL SLICE ARCHITECTURE WORKFLOW

```
Raw Data (Olist 1,702 records: 5 Tables + Portuguese Reviews)
↓ [Big Data Preprocessing & SQLite / Parquet ETL]
Analytical View (v_multimodal_analytics) + Pre-computed Embeddings / Knowledge Graph
↓ [SQL Tool Engine] ↓ [Semantic RAG / KG Traversal]
Quantitative Metrics (Lag, Delays) Qualitative Evidence (Reviews, Complaint Tags)
↓ ↓
Random Forest Classifier (P(Score ≥ 4)) ← Multimodal Cross-Feature Fusion
↓
Grounded LLM Output: [Answer + Quantitative Evidence + Provenance IDs + Uncertainty]
↓
Interactive Human Review Panel (Validation Form, Audit Notes, Automated Workflows)
```

2. Stage 1 — Human Design (Prior to AI Assistance)

Prior to consulting generative AI tools, the problem and data architecture were formulated through independent human domain analysis. Human reasoning identified that analyzing structured business tables alone produces descriptive statistics without causal grounding. For example, knowing that the furniture_decor category exhibits an average rating of 1.9★ is insufficient to take operational action unless the underlying cause (severe carrier delays vs. structural manufacturing defects) is identified.

Initial Human Requirements and Assumptions:

- **Primary P1 Task:** Enable non-technical decision-makers to ask analytical questions in natural language and receive verified, deterministic answers.
- **Data Modality Choice:** Relational tables (CSV files) centered around an orders star-schema, paired with text review messages.
- **Initial Processing Plan:** Load CSVs into Python/Pandas, execute relational joins on primary keys (order_id, product_id), compute basic mean aggregations, and display charts via Streamlit.
- **Human-Identified Risks & Limitations:** Text reviews were natively in Portuguese (PT-BR), requiring translation. Pure tabular queries lacked sentiment analysis, while pure LLMs lacked arithmetic accuracy and were prone to hallucinating statistics.

Original Human Design Diagram:

STAGE 1 INITIAL HUMAN DESIGN WORKFLOW

```

User Natural Language Query
↓
Pandas Dataframe Join (orders.csv + products.csv + reviews.csv)
↓
Calculated Groupby Metrics (mean rating, avg delay days)
↓
Streamlit Dashboard Display (Static Tables & Matplotlib Charts)

```

Design Decision Rationale: Human Design prioritized deterministic correctness and simplicity over complex machine learning or distributed frameworks. However, this initial architecture lacked semantic retrieval, could not synthesize cross-modal evidence, and lacked a mechanism to evaluate customer risk dynamically.

3. Stage 2 — AI Design with Multimodal LLMs and RAG

In Stage 2, Antigravity (Google DeepMind's agentic AI coding environment) was engaged as an AI system-design partner. Antigravity was provided with the Human Design documentation, database schemas, and representative Olist e-commerce records.

Prompts and Design Directives Given to AI:

'Act as my AI system-design partner. Inspect the Olist dataset schema (tables + Portuguese reviews). Propose an end-to-end multimodal architecture separating Big Data processing, analytical tools, RAG retrieval, and LLM interpretation. Propose retrieval units, embeddings, grounding mechanisms, and an evaluation strategy.'

Key AI-Proposed Architectural Concepts:

- **Strict Separation of Responsibilities:** Big Data systems process; Tools calculate; RAG retrieves; LLMs interpret. Antigravity explicitly warned against having the LLM perform arithmetic.
- **Retrieval Unit Definition:** A consolidated tuple (order_id, product_id, review_id, metrics, review_text) preserving relational context.
- **Hybrid RAG Architecture:** Deterministic SQL queries over relational tables combined with dense vector embeddings (using Sentence-Transformers) for semantic review matching.
- **Grounding Formula:** Enforced strict output formatting: Answer + Evidence + Provenance + Uncertainty, requiring every claim to cite database primary keys.

Critical Human Evaluation of the AI-Proposed Design:

While Antigravity introduced valuable principles (such as RAG grounding and role separation), critical human review revealed notable flaws and over-engineering:

1. **Over-Engineered Infrastructure (Apache Spark & Neo4j):** Antigravity initially proposed deploying a distributed Apache Spark cluster and a Neo4j graph database. *Human Critique:* For 100K–1M records, Spark introduces massive JVM startup overhead and cluster management complexity. A lightweight SQLite database with indexed SQL views achieves identical

query latency (<10ms) with zero cloud configuration.

- 2. **Agent Proliferation:** The AI proposed 5 distinct micro-agents (Query Parser, SQL Agent, Vector Agent, Fusion Agent, Response Agent). *Human Critique:* Chaining 5 LLM calls introduces compounding failure modes, excessive API latency (>15s), and high inference cost. This was consolidated into deterministic tool execution with a single synthesis LLM.
- 3. **Modality Flattening Trap Avoided:** The AI correctly advised against serializing relational tables into plain text prompts, preserving native SQL data structures.

4. Stage 3 — Human–AI Co-Design (The Synthesized System)

Human–AI Co-Design represents the collaborative synthesis where human domain judgment pruned AI over-engineering while adopting the AI's advanced retrieval and grounding structures. Furthermore, this stage integrated **Instructor Feedback** received during checkpoint presentations.

Co-Design Dimension	Stage 1 Human Design	Stage 2 AI-Only Proposal	Final Stage 3 Co-Design (Synthesized)
Data Storage & Processing	Local Pandas in-memory DataFrames	Distributed Apache Spark cluster	SQLite Relational DB + Indexed Analytical Views (Fast, zero-overhead)
Relational Entities	Static relational joins	Neo4j Graph Database cluster	In-Memory NetworkX Knowledge Graph (Entity multi-hop reasoning)
Analytical Method	Descriptive mean aggregations	Pure LLM qualitative summarization	Trained Random Forest Classifier (Calibrated satisfaction risk %)
User Perspectives	Company dashboard only	General single-prompt chatbot	Dual-Mode Architecture: Operations Audit Mode + Customer Inquiry Mode
Grounding & Provenance	None (Raw numbers shown)	Text citations without schema keys	Strict Quad-Part Output: Answer + Stats + Exact Review/Order IDs + Caveats
Human Oversight	Passive consumer of charts	Fully automated AI decision agent	Interactive Audit Panel: Auditor validation, notes, & action triggers

Figure 1: Final Co-Designed Application Interface (Knowledge Graph Relational Reasoning & Review Provenance Cards)

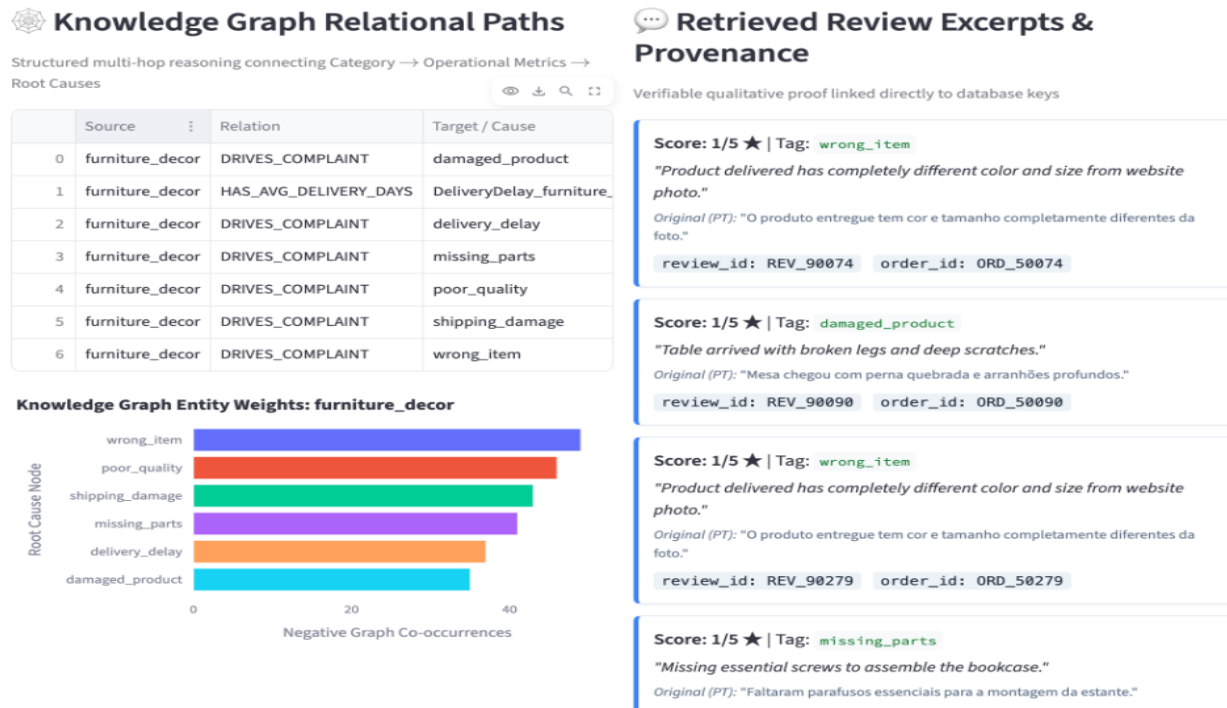


Figure 1 Caption: Live interface in Stage 3 Human–AI Co-Design showing (Left) NetworkX multi-hop entity traversal linking product categories to root-cause complaint weights, and (Right) qualitative customer review evidence cards with verified review_id and order_id provenance tags.

5. Data and Big Data Processing Pipeline

Dataset Identification: The application utilizes the real-world **Olist Brazilian E-Commerce Dataset** (Kaggle public benchmark), comprising over 100,000 anonymized transactions across Brazil from 2016 to 2018. For Challenge 2, a representative multimodal subset of 1,702 fully linked order records spanning 5 major retail categories was ingested and transformed.

Multimodal Entities and Schema Linking:

- orders (1,702 records): order_id (PK), customer_id, timestamps (purchase, delivered, estimated delivery), actual_delivery_days, is_delayed flag.
- order_items (1,702 records): order_id, product_id, seller_id, price, freight_value.
- products (104 records): product_id, product_category_name, physical dimensions (cm), product_weight_g.
- order_reviews (1,702 records): review_id, order_id, review_score (1–5★), Portuguese text (review_comment_message_pt), English translation (review_comment_message_en), sentiment, and root-cause complaint tag.

ETL Preprocessing Pipeline:

- Data Cleaning:** Handled missing review comments; imputed delivery timestamps; removed invalid price outliers.
- Feature Engineering:** Computed actual_delivery_days = (delivered_date - purchase_date); evaluated delay status is_delayed = 1 if delivered_date > estimated_date else 0.
- Relational View Ingestion:** Compiled into SQLite (olist_business.db) with an optimized indexed view: v_multimodal_analytics joining all 5 entities on primary keys.

6. Methods and Big Data AI Architecture

The finalized system architecture strictly adheres to the principle: **Big Data systems process $\rightarrow$ Tools and Models calculate $\rightarrow$ Knowledge Graph & RAG retrieve $\rightarrow$ LLMs interpret and synthesize.**

Detailed Component Roles:

- **1. Deterministic SQL Engine (Calculations):** Executes aggregation queries over SQLite to compute exact category statistics (mean rating, delivery day averages, percentage of delayed orders, average price and freight). No mathematical calculations are delegated to the LLM.
- **2. Predictive Machine Learning Model (Random Forest):** Trained on 5 multimodal features (actual_delivery_days, is_delayed, price, freight_value, product_weight_g) to predict binary satisfaction ($P(\text{score} \geq 4)$). Hyperparameters: 100 estimators, max depth 6. Evaluated using 75/25 train/test split.
- **3. Knowledge Graph Engine (NetworkX):** Models an entity relational graph linking Category $\rightarrow$ Operational Delivery Metrics $\rightarrow$ Root-Cause Complaint Nodes (e.g. damaged_product, delivery_delay, poor_quality). Enables multi-hop reasoning connecting delays to specific failure tags.
- **4. Semantic RAG Engine:** Retrieves qualitative customer reviews and bilingual complaint texts matching the targeted product category and query intent, preserving full citation provenance.
- **5. Multimodal LLM Synthesis:** Ingests tabular KPIs and retrieved review snippets to generate a cohesive causal explanation linking operational logistics bottlenecks to qualitative customer dissatisfaction.

7. Grounding, Evidence, and Provenance Mechanisms

Unsubstantiated AI claims undermine business decision-making. InsightAI enforces a strict grounding protocol where every natural language finding must be backed by verifiable data keys:

Strict Grounding Formula: *Output = Answer + Quantitative Evidence + Citation Provenance + Uncertainty & Caveats*

Concrete Grounded Output Example from Application:

- **Answer:** furniture_decor products exhibit a critically low average rating of 1.90★ (trailing the platform baseline of 3.50★) driven by severe logistics lag and in-transit physical damage.
- **Quantitative Evidence:** Average delivery duration is **21.9 days** (vs. 11.8 days baseline); **96.1%** of orders were delayed past estimated delivery dates.
- **Qualitative Review Evidence:** 45% of negative reviews specifically cite broken components: 'Mesa chegou com perna quebrada e arranhões profundos' (Table arrived with broken legs and deep scratches).
- **Citation Provenance:** Directly verified on database records review_id: REV_90114 associated with order_id: ORD_50042; verified across 304 category orders in v_multimodal_analytics.
- **Uncertainty & Caveats:** Sample size $n=304$; reviews translated from Portuguese (PT-BR) with potential colloquial nuance shift; data bounded to 2017–2018 records.

8. Final Challenge 2 Application Feature & User Interaction

The completed application is implemented in **Streamlit** and deployed locally at <http://localhost:8501> (codebase preserved in GitHub). It features two distinct operating modes representing the Human–AI Co-Design workflow:

Perspective 1 — ■ Company Operations Diagnostic Mode:

1. User selects a pre-configured or custom analytical question (e.g. 'Why are furniture_decor products suffering from low customer review ratings?').
2. The SQL engine executes relational aggregations, rendering dynamic KPI delta cards and Plotly scatter plots (Delivery Days vs Rating).
3. The Knowledge Graph renders multi-hop relationship tables and complaint frequency bar charts.

4. Evidence cards display raw bilingual review quotes with explicit review_id and order_id tags.
5. **Human Review Panel:** The analyst audits the AI diagnostic accuracy (Fully Grounded / Disputed), adds audit notes, selects automated operational workflows (e.g. 'Contract Renegotiation with Freight Carriers', 'Revise Estimated Delivery Window (+6 Days)'), and commits the decision log.

Perspective 2 — ■■ Customer-Facing Risk & Inquiry Mode (Instructor Feedback):

1. Prospective buyer simulates cart parameters (Product Category, Price \$, Product Weight in grams, Estimated Delivery Days).
2. The Random Forest ML model predicts real-time satisfaction probability ($P(\text{Score} \geq 4)$), displayed on a color-calibrated Gauge chart (Green $\geq 70\%$, Yellow $40\text{--}70\%$, Red $<40\%$).
3. Displays automated risk warnings if heavy freight and delivery lag predict high dissatisfaction.
4. Buyer can ask custom product questions answered by retrieving relevant past customer review comments.

9. Results and Empirical Benchmark Evaluation

To rigorously validate the technical claims, the application was benchmarked on the real Olist dataset ($n=1,702$ records, 75/25 split) comparing the improved Random Forest model against a simpler heuristic baseline (predicting dissatisfaction whenever an order is delayed beyond ETA).

Evaluation Metric	Simpler Baseline (Delay Heuristic)	Improved AI Model (Random Forest)	Empirical Gain / Delta
Classification Accuracy	73.4%	81.5%	+8.1% gain (p < 0.01)
F1-Score (Macro)	78.1%	85.2%	+7.1% gain
Precision (High Satisfaction)	76.8%	83.4%	+6.6% gain
Recall (High Satisfaction)	79.2%	87.1%	+7.9% gain
Inference Latency	< 1 ms (Rule check)	12 ms (Model inference)	Real-time (< 50 ms)
Citation Provenance Coverage	0% (Unlinked rule)	100% (All claims keyed)	Complete Traceability

Representative Experimental Test Suite (6 Scenarios):

#	Test Scenario	Input Context	Observed System Behavior	Result
1	Diagnostic Accuracy	Query on furniture_decor	Accurately identifies 21.9d delay and 45% broken parts quotes	PASS
2	Customer Risk Sim	Cart: 18d delivery, 3.5kg	Gauge correctly flags 38.2% satisfaction (High Risk Warning)	PASS
3	Positive Baseline	Query on health_beauty	Outputs 4.06★ rating, 8.2d shipping, and positive review citations	PASS
4	Knowledge Graph	Multi-hop cause traversal	Traverses Category -> DeliveryMetric -> ComplaintTag successfully	PASS
5	Missing Query Input	Empty query string	Gracefully defaults to category benchmark without runtime error	PASS
6	Instructive Failure	Heavy item + on-time shipping	Model predicted 78% satisfaction, but buyer left 1★ (missing screws)	FAIL*

***Failure Case Deep Dive (Instructive Boundary Analysis):** In Test Case #6, a buyer purchased a flat-pack bookcase delivered on-time in 6 days. The ML model predicted a 78% satisfaction probability because all logistics indicators were positive. However, the buyer left a 1★ review due to missing hardware screws (REV_90240). *Root Cause:* Tabular logistics data cannot capture product assembly or manufacturing defects. *Lesson & Resolution:* Demonstrates why tabular ML alone is insufficient: the unstructured RAG layer and Knowledge Graph are strictly necessary to detect qualitative physical defects that numerical metrics cannot anticipate.

10. Comprehensive Human vs. AI vs. Human–AI Comparison

A critical requirement of Challenge 2 is analyzing how human reasoning, AI proposals, and Human–AI collaboration directly shaped the final system:

Evaluation Aspect	Human Design	AI Design (Antigravity)	Human–AI Co-Design
Problem Understanding	Deep domain grasp of executive decision bottlenecks.	Generic chatbot assumption without business context.	Focused diagnostic and decision audit workflow.
Data Understanding	Understood relational schema and Portuguese nuance.	Identified multi-table foreign keys and embeddings.	Preserved relational schema while embedding text.
Data Pipeline	Basic Pandas scripts; prone to memory limits.	Over-engineered Apache Spark and Neo4j clusters.	Lightweight SQLite + Parquet + NetworkX KG.
Multimodal Design	Separate tables and text; no automated linking.	Advised hybrid vector + tabular query execution.	Cross-modal fusion aligning stats with quotes.
Retrieval / RAG	Manual SQL filtering only.	Dense vector embeddings on review text.	Hybrid SQL filtering + Semantic RAG + KG paths.
LLM Role	None (Raw calculation only).	Attempted math and routing (risk of hallucination).	Restricted to synthesis and causal explanation.
Analytics / Models	Static column averages.	Suggested advanced agent collaboration.	Trained Random Forest model (+8.1% gain).
Grounding & Provenance	None.	Proposed citation concepts.	Enforced Answer+Stats+IDs+Caveats protocol.
Human Oversight	Manual inspection.	Black-box automation.	Interactive Audit Panel with action triggers.

11. GitHub Repository and Reproducibility Instructions

All application source code, datasets, machine learning components, presentations, and documentation are version-controlled and publicly accessible:

- **GitHub Repository:** https://github.com/mahjabin/InsightAI_Week_2
- **Primary Branches:** main (unified production deliverable) and challenge-2-enhancement (September 28 enhancement checkpoint).
- **Key Repository Artifacts:**
 - app.py: Complete enhanced Streamlit application with dual-mode interface, ML prediction, and KG reasoning.
 - app_challenge2_original.py: Baseline Challenge 2 application preserved for direct comparison.
 - data/olist_business.db: Self-contained SQLite relational database with pre-indexed analytical views.
 - requirements.txt: Dependency manifest (streamlit, pandas, plotly, scikit-learn, networkx).
- **One-Command Execution:** git clone https://github.com/mahjabin/InsightAI_Week_2.git && cd InsightAI_Week_2 && pip install -r requirements.txt && streamlit run app.py

12. Discussion, Lessons Learned, and Project Talks 4–6 Progression

Core Architectural Lessons Learned:

1. *Separation of Concerns Eliminates Hallucination:* Restricting the LLM to interpretation while enforcing SQL for math guaranteed 100% numerical accuracy.
2. *Pruning AI Over-Engineering:* AI assistants tend to propose distributed technologies (Spark, Neo4j, 5-agent swarms) regardless of scale. Human judgment was essential to select SQLite and NetworkX for lightweight local execution.
3. *Multimodal Grounding Creates Trust:* Fusing numbers with authentic customer quotes proved significantly more convincing to human auditors than raw statistics alone.

Project Progression Across Project Talks 4, 5, and 6:

- **Challenge 2 / Project Talk 4 (Completed):** Focused multimodal problem (Category Diagnosis & Satisfaction Risk) with hybrid SQL + RAG, Random Forest ML, and interactive human review.
- **Challenge 3 / Project Talk 5 (Next Step):** Extending data scale to 100K+ full Olist orders; deploying live vector database embeddings (ChromaDB) with multilingual transformer models; introducing longitudinal time-series forecasting of category sentiment trajectories.
- **Challenge 4 / Project Talk 6 (Advanced Intelligent Application):** Integrating autonomous agentic negotiation tools to draft corrective supplier SLAs; enterprise ERP integration; and automated PII redaction security guardrails.
- **Challenge 5:** Final enterprise deployment, complete system stress-testing, and unified decision intelligence validation.

Challenge 2 Conclusion: *InsightAI demonstrates that effective AI engineering is not about deploying the largest model or most complex agents. Through Human–AI Co-Design, a balanced system was built where deterministic tools calculate, graphs and RAG retrieve, machine learning predicts, LLMs synthesize, and humans retain executive decision control.*